

# PANUWAT SANGKETKIT

## Embedded Systems & Validation Engineer

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### PROFESSIONAL SUMMARY -----

Embedded and control systems engineer with 7+ years of experience in software development, system evaluation, robotics, real-time control, and automotive ECU validation. Experienced in C, Python, MATLAB/Simulink, ESP32 firmware development, sensor and communication integration, circuit and PCB design, and oscilloscope-based prototype testing. Able to investigate system issues from both hardware and software perspectives.

### WORK EXPERIENCE -----

#### System Engineer

Nov 2021 – Present

#### TOTO ELECTRIC INDUSTRY CO., LTD.

Yokohama, Japan

- Develop real-time motion control software for an automated concrete spraying robot using VB.NET and Kawasaki AS language.
- Implement motion-control and sensor-correction functions using IMU and distance-sensor data.
- Develop and verify TCP/IP and UDP communication among PC, PLC, sensors, and robot controllers.
- Led an HMI migration from Keyence to Schneider/Pro-face.

#### Software Engineer

Oct 2019 – Nov 2021

#### EXARS CORPORATION

Aichi, Japan

- Supported automotive ECU evaluation and validation for DENSO-related projects.
- Developed MATLAB/Simulink validation tools using MATLAB m-files.
- Performed ECU and communication checks using CAN and Ethernet, including data analysis with CANoe/CANalyzer and CAPL.
- Installed and evaluated software for a vehicle communication module (DCM).
- Developed ECU test software in C and small checking/automation tools in Python, PowerShell, and VBA.
- Developed a database communication application using C# and MySQL.

#### R&D Engineer

Feb 2018 – Sep 2019

#### Aimagin Co., Ltd.

Samut Prakan, Thailand

- Developed embedded firmware in C for ESP32-based measurement devices.
- Implemented real-time sensor data acquisition and processing for a heavy-metal detection device.
- Designed circuit schematics and PCB layouts and supported prototype assembly and verification.
- Used an oscilloscope during prototype testing and hardware/software debugging.
- Provided technical support for MATLAB/Simulink-based development.

## SKILLS

### Programming:

C, C#, C++, VB.NET, Python, VBA, CAPL

### System Evaluation & Debugging:

ECU validation, functional testing,  
communication checks, log/data analysis

### Embedded Systems:

ESP32(ESP-IDF) firmware, Real-time data  
acquisition and processing, sensor integration

### Communication:

TCP/IP, UDP, CAN, Ethernet, PLC communication

### Operating Systems:

Windows, Linux (Ubuntu, Raspberry Pi), SSH,  
basic terminal operations, package installation

### Hardware & Measurement:

Oscilloscope, Circuit design, PCB design,  
Prototype bring-up

### Tools & Technologies:

MATLAB/Simulink, Git, Visual Studio, VS Code,  
Altium Designer, MySQL, CANoe/CANalyzer

### Control & Automation:

Real-time control, Motion control, PID control,  
Kawasaki AS, PLC/HMI integration

### Languages:

English (Intermediate), Japanese (JLPT N3),  
Thai (Native)

## CERTIFICATIONS

### Industrial Robot Teaching (Teach Pendant) & Safety Training

Mar 2022

Kawasaki Robotics, Japan

## PROJECTS

### Wearable ECG Monitoring Device

2015 – 2017

King Mongkut's University of Technology Thonburi (KMUTT), Thailand

- Developed a low-complexity dual-threshold QRS detection algorithm for real-time wearable ECG devices
- Evaluated algorithm robustness under noise and motion conditions using synthetic and real ECG datasets.
- Designed BLE-based data transmission to an Android device.
- Published research in IEEE conference: <https://ieeexplore.ieee.org/document/8096215>

### Assistive Laundry-Carrying Robot for Elderly

2013

King Mongkut's University of Technology Thonburi (KMUTT), Thailand

- Developed a mobile robot to assist elderly users in carrying laundry.
- Implemented target detection using ultrasonic sensors for distance and direction tracking.
- Integrated digital compass for orientation control.

## ADDITIONAL EXPERIENCE

### Research Intern

Jul 2013 - Dec 2013

### National Electronics and Computer Technology Center (NECTEC)

Pathum Thani, Thailand

### Rehabilitation & Assistive Technology Lab

- Designed PCB circuits and assembled prototype hardware for assistive robotic systems.

## EDUCATION

### Master of Engineering (M.Eng.) in Control Systems & Instrumentation Engineering

2014 – 2018

### Bachelor of Engineering (B.Eng.) in Control Systems & Instrumentation Engineering

2011 – 2014

King Mongkut's University of Technology Thonburi (KMUTT), Thailand